

Varshini Kalakonda (Software Development Engineer @Amazon)

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Professional Summary

Software Development Engineer with **5+ years** of experience in designing and maintaining **scalable backend and frontend systems** using **Java, Spring Boot, AWS, TypeScript, and React.js**, with strong expertise in **automation, system integration, and cloud-native microservices**, focused on **enhancing system reliability** and **supporting critical production environments**; proficient across the **full software development lifecycle**, including **testing and deployment within Agile frameworks**, ensuring **high-quality software delivery**, and recognized for **effective troubleshooting** and **cross-functional collaboration** in **manufacturing and enterprise environments**

Technical Skills

Programming Languages:	Java, JavaScript, TypeScript, C++, C#, Python, SQL
Frameworks & Libraries:	Spring Boot, Spring MVC, Spring Batch, FastAPI, Hibernate, React JS, Angular, Redux
Frontend technologies:	HTML5, CSS3, Bootstrap, Web Architecture, Frontend Development
Web technologies:	RESTful API's, Node.js, JDBC, OAuth2, JWT
Databases:	MySQL, DynamoDB, MongoDB, Oracle, PostgreSQL, Redis
Cloud services:	AWS (Lambda, Fargate, S3, EC2, IAM, SQS, SNS, Step Function, CloudWatch) and Azure
DevOps and CI/CD:	Docker, Kubernetes, Jenkins, Git, Maven, Gradle, Bitbucket, Ansible, XLR, Terraform
Testing and debugging:	Junit, Mockito, Postman, Selenium, Sonar, Crucible, TestNG
Messaging and streaming:	Apache Kafka, GraphQL
Software development:	Object-oriented design, System Design, Web Services, Agile, Algorithms, Data Structure
AI tools:	LLM-based workflows, Prompt Engineering, AI-assisted development (Kiro)

Professional Experience

Software Development Engineer, Amazon, Austin, USA

Jul 2023 - present

- Architected and deployed a **multi-region event-driven system** using **Java, Spring Boot, Kinesis, and DynamoDB Streams**, improving message delivery reliability and reducing latency by **65%** across continents.
- Designed and deployed **Python-based microservices (FastAPI)** to orchestrate **LLM-driven workflows**, enabling automated SOP generation, simulations, and code review, reducing engineering effort by **40%**
- Designed a **microservice-based payment orchestration layer** leveraging **AWS Step Functions and Lambda**, simplifying transaction workflows and reducing code complexity by **30%**.
- Implemented **idempotent message handling** across distributed services using **DynamoDB transactional writes**, ensuring fault tolerance and eliminating duplicate processing.
- Containerized legacy services using **Docker and AWS ECS Fargate**, enabling zero-downtime deployments and reducing release cycle time by **70%**.
- Developed a **self-healing job retry framework** using **AWS Step Functions** and **DLQ redrives**, reducing manual ticket volume by **85%**.
- Automated **data consistency checks** between source and target DynamoDB tables using **AWS Glue and Athena**, catching synchronization issues before production impact.
- Built a **CI/CD validation pipeline** integrating **Gradle, Code Build, and SonarQube**, enforcing code quality gates and cutting post-deploy defects by **50%**.
- Implemented a **real-time metrics ingestion service** using **Kinesis Firehose, Lambda, and OpenSearch**, providing near-instant dashboards for business KPIs.
- Designed a **data archival strategy** using **S3 lifecycle policies** and **Glacier Deep Archive**, cutting storage costs by **35%** without losing retrieval flexibility.
- Built **custom DynamoDB query optimizations** for analytics workloads using GSIs and batching, reducing scan latency by **78%**.
- Developed a **React-based configuration management dashboard** integrated with backend APIs, improving operational visibility and reducing dependency on DevOps teams.
- Designed a **role-based UI rendering system** aligned with backend access controls, enhancing security and user experience consistency.
- Implemented **service-to-service authentication** using **AWS IAM roles and SigV4 signing**, eliminating hardcoded credentials and improving compliance.
- Integrated **AWS KMS encryption** for PII data storage and transmission, aligning with GDPR and ISO 27001 standards.

- Built an **anomaly detection mechanism** using CloudWatch metrics and SNS alerts to flag suspicious API traffic, proactively mitigating potential threats.
- Led a cross-functional initiative to deprecate 10+ legacy APIs, coordinating across 4 teams and reducing operational maintenance by **60%**.
- Mentored 3 junior engineers through their first on-call rotation, improving incident response confidence and reducing MTTR by **40%**.

Senior Software Engineer, Infosys Technologies, Hyderabad, Telangana

Jul 2019 – Dec 2021

- Worked in Live Web Development for **Syngenta**, contributing to end-to-end feature development using **React JS (frontend)** and **Java/Spring Boot (backend)** within an Agile environment.
- Developed and maintained scalable backend services using **Java 8/11**, **Spring Boot**, and **RESTful APIs**, improving system performance by **40%**.
- Built reusable service modules and automation scripts, **reducing development effort by 25%** across multiple projects.
- Implemented data persistence layers with **Hibernate/JPA** and optimized SQL queries for faster database interactions (**MySQL/PostgreSQL**).
- Refactored legacy components and optimized multithreading and caching mechanisms, **reducing response times by 50%**.
- Containerized applications using **Docker** and orchestrated deployments with **Kubernetes** to improve scalability and reliability.
- Integrated CI/CD pipelines using **Azure DevOps**, automating build, test, and deployment workflows and **reducing release time by 30%**.
- Deployed and monitored microservices using **Azure App Service**, **Azure Functions**, and **Azure Application Insights** to enhance system observability.
- Wrote comprehensive **JUnit** and **Mockito** test cases, achieving **95%+ code coverage** and **reducing production bugs by 60%**.
- Prepared regulatory and compliance documentation related to safety and software integrity.
- Collaborated with cross-functional teams to troubleshoot and resolve complex technical issues efficiently.

Education and Training

Master of Science in Computer Science

Jan 2022 – May 2023

Texas A&M University, Corpus Christi, TX, US.

GPA: 3.75

Coursework: Algorithms, Data Structures, Cloud Computing, Distributed Systems, Software Engineering

Bachelor of Technology in Electronics and Communication Engineering

Aug 2015 – May 2019

Kakatiya Institute of Technology and Science, Warangal, Telangana, India

GPA: 3.85

ACHIEVEMENTS AND CERTIFICATIONS

- Certified in Data Structures and Algorithms on Udemy, demonstrating strong problem-solving skills
- Solved 400+ coding challenges across HackerRank, LeetCode, and GeeksforGeeks.
- Certified in Java, SQL, and Problem Solving by HackerRank and AWS.
- Microsoft Azure Fundamentals AZ-900 Certification by clearing assessment.